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A Cross-Modal *Bayesian Reward Filter* for Robust Reinforcement Learning under Noisy Human Feedback

POSTER / ML · RL

BRF

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01 Abstract

> *Can a reinforcement-learning agent limit the effects of biased or noisy human feedback by using sparse environment rewards as a probabilistic reality check through online Bayesian filtering?*

Standard **RLHF** assumes that feedback from human supervisors is the ground truth, leaving agents vulnerable to bias, fatigue, and adversarial labelers [1]. We introduce a **Cross-Modal Bayesian Reward Filter (BRF)** that fuses human preference signals with sparse, verifiable environment rewards. The filter maintains a posterior **Trust Score w** over the human channel from the running likelihood of agreement between the two modalities, and uses it to down-weight or reject corrupted feedback online — without retraining or hand-tuning a uncertainty penalty.

We evaluate on **MiniGrid** [3] with synthetic adversarial, fatigue, distance-noise, and realistic human models, comparing PPO and DQN agents against a sparse-only baseline and naive RLHF. **BRF recovers the optimal policy under realistic noise (success rate 1.0 vs. 0.667 naive)** and matches the sparse baseline against pure adversaries — suggesting Bayesian arbitration is a low-cost safety layer for any preference-based pipeline.

02 Introduction & Objectives

LANDSCAPE OF CURRENT APPROACHES

METHOD	MECHANISM	LIMITATION
Standard RLHF Christiano et al., 2017 [1]	Trains a reward model on pairwise human preferences; PPO optimizes against it.	Treats human labels as ground truth — collapses under bias.
Uncertainty-Penalized RL Coste et al., 2024 [2]	Reward-model ensemble; penalizes disagreement to slow over-optimization.	Requires $k\times$compute and a fixed penalty schedule.
Bayesian RL Ghavamzadeh et al., 2015	Maintains posterior over MDP / value function; principled exploration.	Rarely applied *between* reward modalities.

Gap. Existing work either trusts the human *or* guards the reward model — none arbitrates between two reward streams in real time. The BRF asks the environment to vote whenever the human speaks.

OBJECTIVES

- Formalize a **cross-modal likelihood** linking human feedback to a sparse environment reward.
- Implement an online Bayesian update that yields a per-step **Trust Score $w \in [0,1]$**.
- Quantify robustness across **six adversarial human models** on three MiniGrid tasks.
- Compare against sparse-only and naive RLHF using **PPO** [5] and **DQN** [6] agents.

03 Methodology

THE BAYESIAN REWARD FILTER

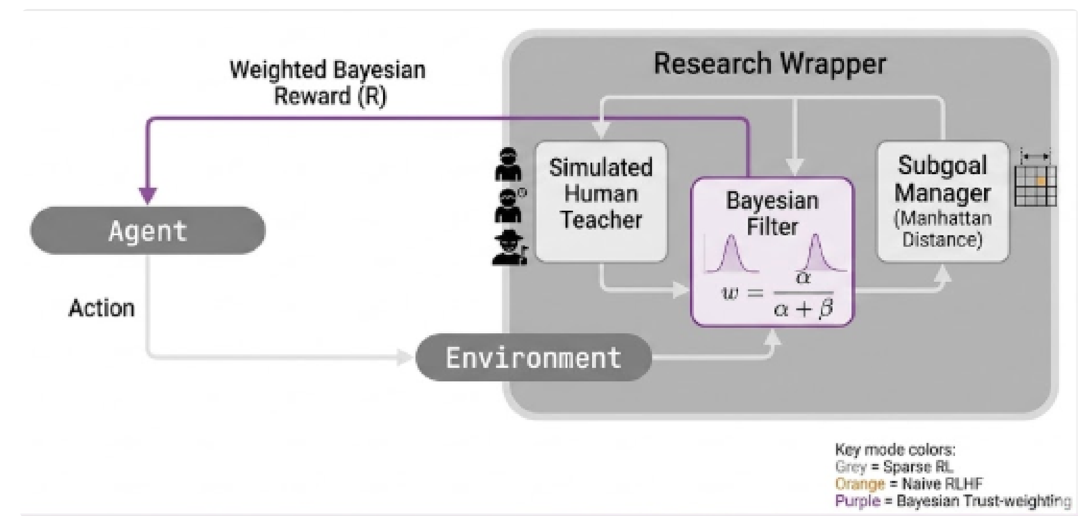


Fig. 1. *The BRF sits inside the research wrapper: a* **Bayesian Filter** *with Beta posterior $w = \alpha/(\alpha+\beta)$ arbitrates the simulated human teacher against a Manhattan-distance subgoal manager, returning a weighted reward R to the agent.*

Each transition *(s,a,s′)* yields a sparse environment reward and a (potentially corrupted) human label. Their joint likelihood updates the posterior over *w*; the policy optimizes the weighted reward above using PPO or DQN.

ENVIRONMENT REWARD — DENSE POTENTIAL SHAPING

$$r_{env} = 1 - 0.9 \cdot (\text{steps} / \text{max\_steps}) + \gamma \cdot \Phi(s') - \Phi(s)$$

where $\Phi(s) = -\text{Manhattan(agent, goal)}$; staged for DoorKey.

HYPERPARAMETERS

STEPS / RUN	15,000
PPO EPOCHS	10
LEARN RATE	3×10^{-4}
OPTIMIZER	Adam
EVAL EPS.	20
PRIOR	Beta(1,1)

SYNTHETIC HUMANS

Stochastic	Fixed-rate label flip baseline.
Adversarial	Always lies — worst case.
DistanceNoise	Confusion grows with distance to goal.
Fatigue	Noise increases over training time.
Realistic	Distance + fatigue + base noise.

ENVIRONMENTS

Empty-8x8-v0	Reach green tile; $\Phi = -$Manhattan.
LavaGapS7-v0	Avoid lava strip; Φ adds adjacency penalty.
DoorKey-8x8-v0	Pickup → unlock → reach; staged Φ.

All MiniGrid [3]; preference protocol mirrors Uni-RLHF [4].

04 Results

EVAL SUCCESS RATE · PPO · MINIGRID-EMPTY-8×8

HUMAN MODEL	SPARSE ONLY	NAIVE RLHF	BRF (OURS)
Adversarial	0.000	0.000	0.000
DistanceNoise	0.333	0.333	0.333
Fatigue	0.333	0.333	0.667
Realistic	0.333	0.667	1.000
Stochastic Hi	0.000	0.000	0.000
Stochastic Lo	0.000	0.667	0.667

Table 1. *Reported as fraction of 20 evaluation episodes that reach the goal under each reward source. More complex environments (LavaGap, DoorKey) proved unsolvable by all reward systems at this budget — the sparse environment signal alone is too thin to bootstrap.*

CROSS-ENVIRONMENT HEATMAP

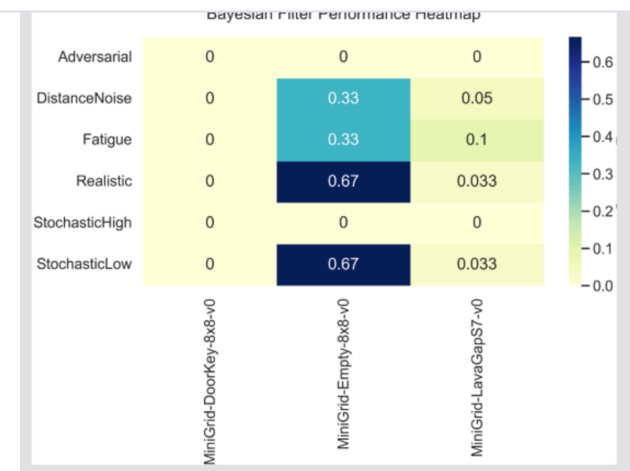
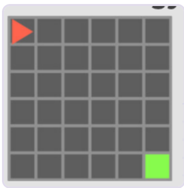


Fig. 2. *BRF success across MiniGrid tasks. Empty–8×8 is solvable; LavaGap collapses for most humans; DoorKey unsolvable at this budget.*



▲ REALISTIC NOISE

+33.3 pp

Full recovery vs. naive RLHF (1.000 vs 0.667).

◆ ADVERSARIAL

w→0

Filter rejects the liar; falls back to sparse reward.

05 Discussion & Future Work

- **Bayesian arbitration is essentially free.** One Beta posterior per channel, no ensembling — yet matches or beats naive RLHF in every non-degenerate setting.
- **Next:** pair BRF with reward shaping for sparse tasks (LavaGap, DoorKey); test Uni-RLHF crowd labels [4] and reward-hacking attacks.

06 · REFERENCES

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07 · ACKNOWLEDGEMENTS

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